ML SYSTEMS

Federated Learning Visual Guide

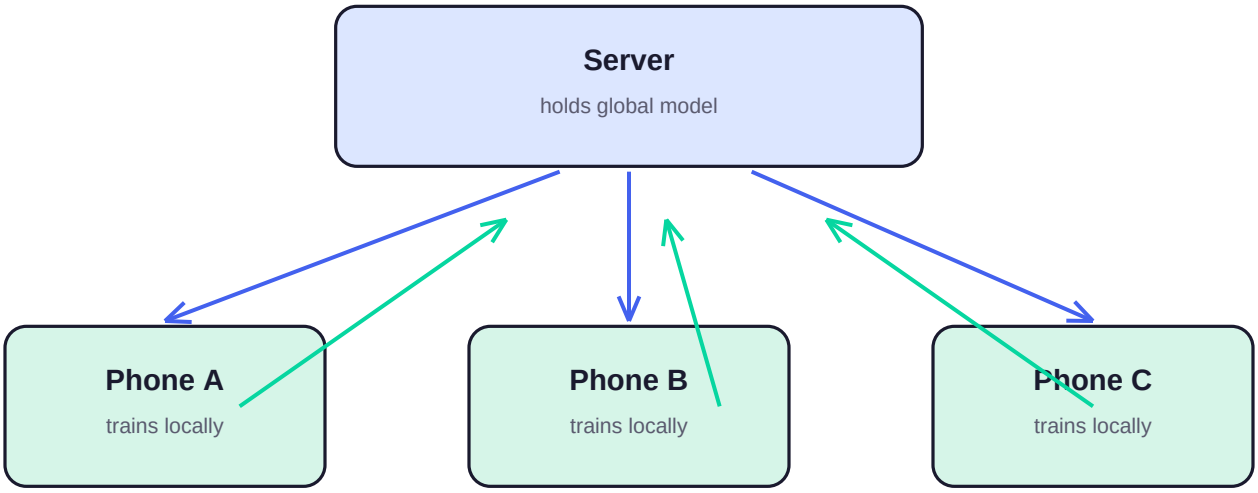
Training without collecting data:
rounds, secure aggregation, non-IID challenges.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

Federated Rounds

The model travels; the data stays home.



Round: server sends model -> devices train on local data -> server averages updates (FedAvg) -> repeat.

Raw data never leaves the device. Only model updates travel — the core privacy promise.

STEP 2/5

Secure Aggregation

The server learns the sum, never the parts.

Masked update

mask + update + random

Masked update

mask + update + random

Server

sees only the aggregation. Masks cancel in total.

Cryptographic masking means even a curious server can't reconstruct any single device's update.

STEP 3/5

The Non-IID Challenge

Every device's data looks different.

Non-IID reality

one user types in Spanish, another in emoji. Skewed labels.

IID fantasy

every device sees the same distribution. Algorithms assume this.

Non-IID data slows convergence and can diverge FedAvg. Fixes: proximal terms (FedProx), personalization layers, more rounds.

STEP 4/5

Privacy Techniques

Defense in depth for updates.

- Differential privacy: add calibrated noise to updates so no single user is identifiable.
- Secure aggregation: cryptography hides individual updates from the server.
- Update clipping: bound each device's influence — also blunts poisoning attacks.
- On-device personalization: keep a private head locally; share only the base model.

Model updates can leak training data (inversion attacks) — 'we never saw the data' is not the whole story.

STEP 5/5

When Federated Fits

The right tool for sensitive, distributed data.

- Keyboards, voice assistants, health apps: data users won't (or can't) upload.
- Hospitals collaborating without sharing patient records across institutions.
- Requirements: many clients, intermittent connectivity, tolerance for slow convergence.
- Skip it when data CAN be centralized — federated is always harder than centralized training.
- Watch for: stragglers, dropped devices, and adversarial clients poisoning the global model.

STEP ✓/✓

Key Takeaways

Federated Learning — Remember

- Federated rounds: server broadcasts model, devices train locally, server averages (FedAvg).
- Data stays on device; only updates travel — with secure aggregation hiding individuals.
- Non-IID data is the core challenge: skewed clients slow or break convergence.
- Add differential privacy and clipping; updates can leak data without them.
- Use when data can't be centralized; prefer centralized training otherwise.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev